Simultaneous equations models

1 A cat moves down a path at 8 m/s. A dog begins moving down the same path from the same point at a speed of 12 m/s, but starts 2 seconds later.

a Graph the distance travelled by each animal, measuring the time from when the cat starts running.

b How long will it be before the dog catches up with the cat?

c How far from the initial point will this be?

2 A family owns two businesses, EcoLine and Helios. These businesses made a combined profit of $\$12$ million in the previous financial year, with Helios making 3 times as much profit as EcoLine.
Let x represent the profit (in millions of dollars) of EcoLine, and y be the profit (in millions of dollars) of Helios.

a Use the fact that the two businesses made a combined profit of $\$12$ million to set up an equation involving x and y.

b Use the fact that Helios made 3 times as much as EcoLine to set up another equation relating x and y.

c Graph the two equations on the same set of axes.

d Use the graph to find EcoLine's profit.

e Use the graph to find Helios' profit.

3 A rectangular zone is to be 2 cm longer than it is wide, with a total perimeter of 20 cm.
Let y represent the length of the rectangle and x represent the width.

a Complete the following two equations that represent the information.

i $y = \square + \square$ **ii** $2x + \square = 20$

b Graph the two equations on the same set of axes.

c Use the graph to find the length and width of the rectangle.

i Length   **ii** Width

4 A mother is currently 6 times her son's age. In 4 years time, she will be 4 times her son's age.
Let x and y be the present ages of the son and mother respectively.

a Use the fact that the mother is currently 6 times her son's age to set up an equation relating x and y, where y is the subject of the equation.

b Use the fact that the mother will be 4 times her son's age in 4 years time to set up another equation relating x and y. Write the equation in the form $y = mx + b$.

c Solve the system of linear equations using technology.

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5 Toby's piggy bank contains only 5c and 10[redacted]s. He knows that there are 53 coins in the piggy bank, and that the total value is $\$3.50$.
Let x and y be the number of 5c and 10c coins respectively. We will create two equations then solve them simultaneously to find the number of coins of each type.

a Use the fact that the total number of coins is 53 to set up an equation relating x and y.

b Use the fact that the total amount of coins are worth $\$3.50$ to set up another equation relating x and y.

c Solve the system of linear equations by either graphing or by using technology.

6 For two numbers x and y:

- Six times the first number is added to the second number to get 71.
- The difference between eight times the first number and the second number is 83.

a Set up two equations relating x and y. Use x as the first number and y as the second.

i Sum equation

ii Difference equation

b Solve the system of linear equations in part (a), using appropriate technology.

7 There are 28 members in a monkey troupe, and the females outnumber the males by 4.
Given this information, we want to find the number of each gender in the troupe.
Let x be the number of male monkeys and y be the number of female monkeys in the troupe.

a Use the fact that the females outnumber the males by 4 to set up an equation with y as the subject.

b Use the fact that there are a total of 28 members in the troupe to form another equation relating x and y. Write the equation in a form where the terms involving x and y are on one side, and the constant term is on the other.

c Solve the system of linear equations using technology.

8 A gym offers aerobics classes where non-members pay $\$3$ per class and members pay a $\$4$ fee plus an additional $\$2$ per class. The monthly cost, y, of taking x classes can be modelled by the linear system:

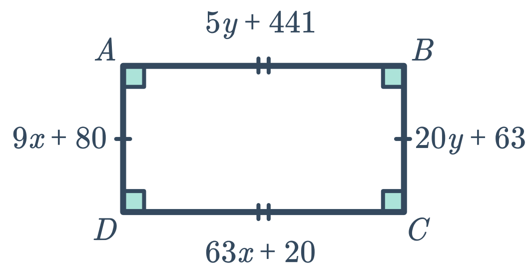
- Non-members: $y = 3x$
- Members: $y = 2x + 4$

a Graph the two equations on the same set of axes.

b State the values of x and y which satisfy both equations.

c What do the coordinates of the solution mean?

9 Consider the rectangle $ABCD$ given:



a Use the fact that $AB = CD$ to set up Equation 1.

b Use the fact that $AD = BC$ to set up Equation 2.

c Solve the system of linear equations using technology.

d Find the length of the side CD.

e Find the length of the side AD.

10 The existing incandescent light bulbs in Buzz's home cost $\$6.00$ per month to operate. He is considering switching to new energy-efficient fluorescent bulbs, which will cost $\$4.00$ per month to operate. It will cost him $\$4.00$ to purchase the new bulbs.

a Write an equation for the cost c of using the old incandescent bulbs for m months.

b Write an equation for the cost c of using the new energy-efficient fluorescent bulbs for m months.

c Graph the two equations on the same set of axes.

d Buzz decides to replace the old incandescent bulbs today and goes to the store to buy them. How many whole months from now will he start saving money overall?

11 Two twins, Amelia and Xavier, play soccer together. The twins have scored a combined total of 34 goals so far and Amelia has scored 8 more goals than Xavier.
Let x be the number of goals scored by Amelia, and y be the number of goals scored by Xavier.

a Use the fact that the twins have scored a combined total of 34 goals to set up an equation involving x and y.

b Use the fact that Amelia has scored 8 more goals than Xavier to set up another equation involving x and y.

c Graph the two equations on the same set of axes.

d Use the graph to find the number of goals Amelia has scored this season.

e Use the graph to find the number of goals Xavier has scored this season.

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12 A clothing manufacturer is deciding whether to employ people or to purchase machinery to manufacture their line of t-shirts. After conducting some research, they discover that the cost of employing people to make the clothing is $y = 400 + 60x$, where y is the cost and x is the number of t-shirts to be made, while the cost of using machinery (which includes the cost of purchasing the machines) is $y = 1000 + 30x$.

a Graph the two cost functions on the same set of axes.

b Find the value of x at which it will cost the same whether the t-shirts are made by people or by machines.

c Find the range of values of x for which it will be more cost efficient to use machines to manufacture the t-shirts.

d Find the range of values of x at which it will be more cost efficient to employ people to manufacture the t-shirts.

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